



My Robot Cheat Sheet

Tell the mBot2 what to do with Python — it moves, glows, and beeps!



★ ALWAYS START LIKE THIS

```
from mbot2 import MBot2

bot = MBot2()           # wakes up & connects to the robot

# 🖱️ put your commands here! (see the cards below)

bot.disconnect()        # say goodbye at the end
```

Move

Speed is 0–100. Add seconds and it stops by itself.

```
bot.forward(50, 2)  drive ahead for 2 seconds
bot.backward(50, 2) drive backwards
bot.turn_left(50, 1) spin left
bot.turn_right(50, 1) spin right
bot.turn(90)        turn exactly 90° (+ right, – left)
bot.stop()          stop the wheels now
```

Lights

Mix red, green & blue (each 0–255). See the mixer below!

```
bot.led(0, 255, 0)  all 5 lights turn green
bot.led(255, 0, 0, which=1) just light #1 (1–5) turns red
bot.led_off()        lights off
bot.led_effect("rainbow") a rainbow light show 🌈
```

Screen & Wait

Words on the robot's little screen.

```
bot.show("Hi!")  show text on the screen
bot.sleep(1)     wait 1 second, then keep going
```

Sensors

Ask the robot a question — it gives an answer back.

```
bot.distance()  how far away is a wall? (cm)
bot.battery()   battery left (0–100)
bot.brightness() how bright is the room?
bot.loudness()  how noisy is it?
bot.button("a") is button A being pressed?
```

Sound

Bigger number = higher squeaky note.

```
bot.beep(880)  play a high beep
bot.beep(220)  play a low beep
bot.play_note(60) play a piano note (0–132)
bot.volume(80) how loud (0–100)
```

Mix your own light color

`bot.led( red , green , blue )`

```
bot.led(255, 0, 0)  red
bot.led(0, 255, 0)  green
bot.led(0, 0, 255)  blue
bot.led(255, 255, 0) yellow
bot.led(200, 0, 255) purple
bot.led(0, 255, 255) aqua
bot.led(255, 255, 255) white
```

Good to know

-  Small speed = slow & gentle. Big speed = zoomy! Try `20` then `80`.
-  The second number is **time**. `forward(50, 3)` drives for 3 seconds.
-  `turn(90)` goes right, `turn(-90)` goes left.
-  Give a command a friend — put `bot.sleep(1)` between them to pause.
-  Robot gone wild? Turn it off, or run `bot.stop()`.

🚀 TRY THIS WHOLE PROGRAM!

```
from mbot2 import MBot2

bot = MBot2()

bot.show("Let's go!")
bot.led(200, 0, 255)  # glow purple
bot.forward(40, 2)    # drive for 2 seconds
bot.turn(180)         # spin around
bot.led_effect("rainbow") # rainbow party 🌈
bot.beep(880)         # happy beep!

bot.disconnect()
```